Random Variables and Distributions

Unit 8: Putting a Number on an Uncertain Thing

Stat 220 | Statistics for Data Science

Where this fits

- Unit 7 dealt in events: did the customer buy, is the test positive, did the server fail.
- Now we attach a **number** to the outcome. Revenue, wait time, defects, dollars lost.
- Once outcomes are numbers we can average them, add them up, and compare decisions, which is where this becomes useful to somebody with a budget.

Principle: a random variable is a rule, not a value

It is not "the number that came up." It is the whole recipe: which numbers are possible and how often each one occurs.

What this unit gives you

Things to know

- expectation as a long-run average, and its linearity
- variance and standard deviation, in the right units
- what happens to the mean and spread when you add things up
- which standard distribution matches which real-world story
- why plugging an average into a curved calculation is wrong

Mistakes to stop making

- reporting a mean with no measure of spread
- treating the mean as typical when the data is skewed
- computing the answer at average inputs instead of averaging the answers
- adding variances of things that are not independent
- picking a distribution by eyeballing a histogram

Random variables and expectation

A **random variable** is not "the number that came up." It is the whole recipe: which numbers are possible and how often each occurs.

$$E[X] = \sum_x x \cdot P(X = x)$$

- The expectation is the average you would see over a very large number of repetitions.
- It need not be a possible value: the expected number of children per family is 2.3, and no family has 2.3 children.

Principle: expectation answers "in the long run," not "this time"

It is the right tool for decisions you make repeatedly, and a dangerous one for a decision you make once with everything on the line.

Linearity: the most useful fact in this course

For *any* random variables, independent or not:

$$E[aX + b] = aE[X] + b, \qquad E[X + Y] = E[X] + E[Y].$$

- No independence required. That is why expectations are so pleasant to work with.
- **Worked example.** A call center takes 400 calls a day. Each caller buys with probability 0.06, spending \$85. Write $R = \sum_{i=1}^{400} 85\, I_i$ where I_i is 1 if caller i buys. Then $E[R] = 400 \times 85 \times 0.06 = \$2{,}040$.
- We never needed the distribution of daily revenue. Linearity gave us the mean without it.

Variance: how far things swing

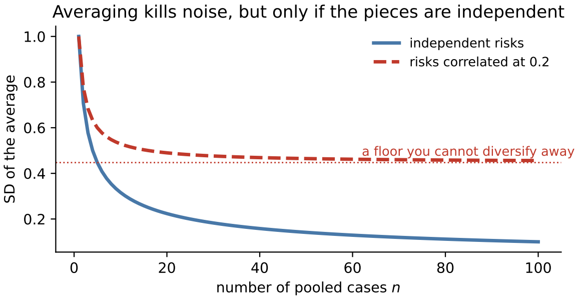
$$\mathrm{Var}(X) = E[(X - E[X])^2], \qquad \mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}.$$

- Variance is in squared units (dollars squared), which nobody can interpret. Report the **standard deviation**, which is in the original units.
- $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$: shifting does nothing, scaling squares.
- For **independent** X and Y: $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$. Variances add, so standard deviations grow like $\sqrt{n}$, not n.

Trap: the independence clause, again

"Variances add" is not a law of nature. It is a reward for independence, and it is the same assumption that failed in 2008.

Why averaging works, and when it stops



The SD of an average of n independent cases falls like $1/\sqrt{n}$.

- Four times the data, half the noise. That is the entire economics of sample size, and you met it in Unit 1 as $\sigma/\sqrt{n}$.
- If the cases are correlated, the curve flattens onto a floor you cannot buy your way below.

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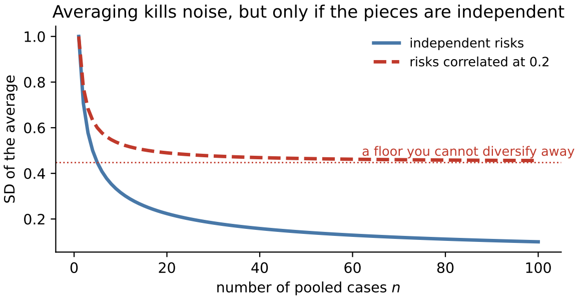

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This one picture is insurance, portfolio diversification, and A/B testing, and also why a pandemic or a market crash breaks all three at once.

Your turn #1

Your turn: same average, different animal

Two ad campaigns each have an expected return of \$50,000.

- Campaign A returns somewhere between \$40k and \$60k.
- Campaign B has a 90% chance of \$0 and a 10% chance of \$500k.

With a neighbor: which do you recommend, and what do you need to know about the company first?

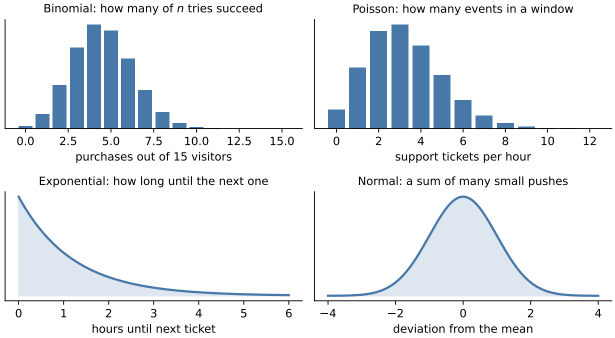
Your turn #1: answer

- There is no answer without context. The real questions are **how many times you get to play** and **what a zero costs you**.
- A firm running 50 campaigns a year should take B. The averaging in the last picture turns its variance into a reliable return.
- A startup that must make payroll next month should take A. One draw of $0 ends the company, and there is no long run for a company that no longer exists.

Principle: expected value assumes you survive to repeat

Maximize expected value across many bets you can absorb. For a single bet you cannot recover from, look at the downside instead.

Pick the distribution by the mechanism



Each shape comes from a *story* about how the data is generated. Learn the four stories and the formulas follow.

The four stories worth memorizing

- **Bernoulli / Binomial:** a fixed number of independent yes-or-no tries, each with the same chance. "How many of 500 visitors converted?" $E[X] = np$, $\mathrm{Var}(X) = np(1-p)$.
- **Poisson:** events arriving at a steady rate in a window of time or space. "How many tickets this hour?" Mean and variance are both λ.
- **Exponential:** the waiting time *until* the next such event.
- **Normal:** anything built from many small independent contributions added together. Unit 3 is entirely about why this one shows up everywhere.

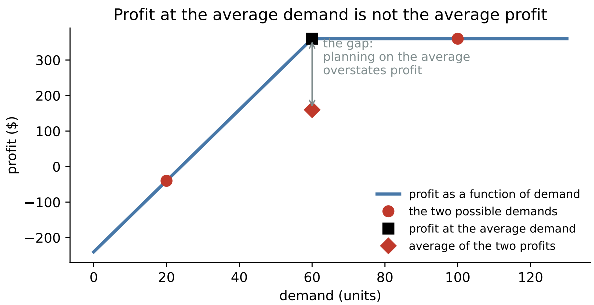
Choosing a model in practice

- Ask what generated the number, not what the histogram looks like.
- Counts of successes out of a known number of tries: binomial. Counts with no natural ceiling, arriving over time: Poisson.
- Waiting times, durations, and time-to-failure: exponential-ish, and almost always right-skewed.
- Money is almost never normal. Revenue per customer, income, and claim sizes are skewed with a long right tail.

Trap: a check worth running

Poisson forces the variance to equal the mean. Real count data is usually *over*-dispersed, with variance well above the mean, because the underlying rate itself varies (Mondays, launches, outages). Check that before trusting a Poisson.

Averages plugged into nonlinear things



Capacity is 60 units, each sold for \$10, each unit of capacity costs \$4. Demand is 20 or 100, equally likely.

- Profit at the *average* demand of 60: **\$360**.
- *Average* profit across the two demands: **\$160**.

The plan built on the average overstates profit by more than double.

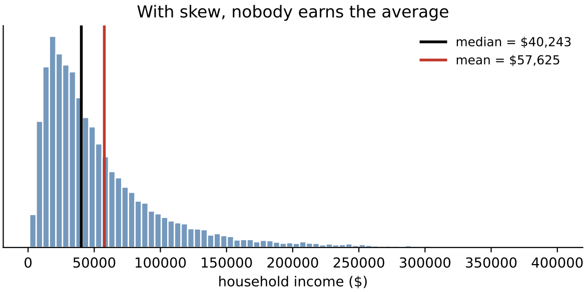
Why the gap appears

- Whenever the quantity you care about is a **curved** function of an uncertain input, $E[f(X)] \neq f(E[X])$.
- Caps, floors, thresholds, penalties, stockouts, and SLA cliffs are all curves. Business is made of them.
- "A statistician drowned crossing a river of average depth three feet."

Trap: planning on the average

Any plan built by replacing every uncertain input with its mean will be systematically wrong, and usually optimistic, because it ignores the bad tail where the penalties live.

The mean is not the typical case



With right-skewed data the mean sits well above the median, dragged up by the tail.

- “Average revenue per user” in a business where 2% of users are whales describes nobody.
- Report the median, the mean, and something about the tail. They answer different questions.

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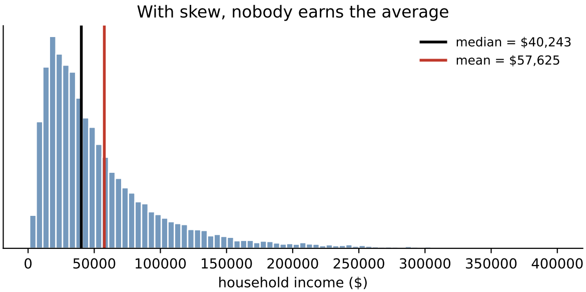

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Rule of thumb: when the mean and median differ a lot, someone is about to be misled.

Modeling an uncertain quantity, in order

1. **Define the variable in words and units.** "Tickets per hour," not "tickets."
2. **Tell the generating story.** Fixed tries? Arrivals over time? A waiting time? A sum of many small pieces?
3. **Pick the distribution the story implies**, and write down what it assumes.
4. **Get the mean and the SD**, and sanity-check both against reality.
5. **Check the assumption that breaks first**, usually independence or a constant rate.
6. **Compute the decision quantity by simulation**, never by plugging in averages.

What you should be able to do with software

Nobody is asking you to memorize syntax. You are expected to know *which* procedure the question calls for and what the output means.

You should be able to	**What you would reach for**
Mean and SD of a simulated quantity	`.mean()`, `.std(ddof=1)`
Exact binomial probabilities and tails	`stats.binom.pmf`, `.sf`
Counts over a window	`stats.poisson.pmf`, `.sf`
Waiting times	`stats.expon`
Check whether counts are over-dispersed	compare the sample variance to the mean
Expected profit or cost under uncertainty	simulate the scenarios, then average
Find the best decision, not the average one	sweep the choice and take the maximum

If you cannot say what the output means, producing it is worth nothing.

The traps, all in one place

Trap: the ones that bite most often

- Reporting a mean with no standard deviation or range.
- Calling the mean “typical” when the data is skewed.
- Computing the answer at average inputs instead of averaging the answers.
- Adding variances for things that share a driver.
- Choosing a distribution from a histogram instead of a mechanism.
- Maximizing expected value on a one-shot bet you cannot survive losing.

Ten questions to be able to answer

1. Define expected value and give a case where it is the wrong thing to maximize.
2. Two projects have equal expected return. What else do you want to know?
3. Why does the SD of an average shrink like $1/\sqrt{n}$, and when does it not?
4. Our average customer spends $80. What is wrong with planning around that?
5. Why is profit at average demand not the same as average profit?
6. Which distribution would you use for tickets per hour, and what does it assume?
7. Your count data has variance three times its mean. What does that tell you?
8. How would you staff when understaffing and overstaffing cost different amounts?
9. Explain to a manager why the median and mean of our revenue differ.
10. When would you simulate instead of using a formula?

Mastery checklist

You have this unit when you can, from memory:

- define a random variable, its expectation, and its standard deviation
- use linearity to break a total into pieces
- say what happens to the mean and SD when independent things are summed or averaged
- match binomial, Poisson, exponential, and normal to their generating stories
- explain the flaw of averages with a concrete example
- argue when expected value is and is not the right objective

Where we go next

- We just saw that averages of many independent pieces get predictably tighter, like $1/\sqrt{n}$.
- **Unit 9** explains why, and shows they also take on a specific *shape*. That result is what finally justifies every standard error you have used since Unit 1.

Principle: the habit to keep

A number without a spread is a guess. The mean says where, the SD says how much to hedge.